SOMAIYA VIDYAVIHAR

K J Somaiya Institute of Technology
An Autonomous Institute Permanently Affiliated to the University of Mumbai

DEPARTMENT OF INFORMATION TECHNOLOGY

Synopsis Report on Mini Project

Minor in Fundamentals of Human Physiology
and Biomedical Engineering

CholesterolWise

LDL Cholesterol Analyzer
Interactive Analysis and PCSK9 Awareness

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Certificate

This is to certify that the following students of Third Year Engineering, Division B, Department of Information Technology, have prepared the mini project synopsis titled **CholesterolWise - LDL Cholesterol Analyzer** for the Minor in Fundamentals of Human Physiology and Biomedical Engineering during the academic year 2026-2027 under the guidance of **Dr. Reena Lokare**.

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This synopsis presents the problem, objectives, literature survey, system design, implementation methodology, software testing results and future scope of the mini project.

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Date of Examination _____

Signature of Internal Examiner

Signature of External Examiner

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We acknowledge the Department of Information Technology at **K J Somaiya Institute of Technology** for providing the academic setting for this work. The minor course encouraged us to connect a biological mechanism with the design and implementation of an interactive educational website.

We acknowledge the researchers whose published studies on PCSK9, LDL receptors and cholesterol regulation informed the educational content. We also acknowledge the National Library of Medicine and the National Heart, Lung, and Blood Institute for their public health information and reference material.

The official documentation for React, FastAPI, React Three Fiber and Chart.js supported the technical implementation. These resources helped us organize the interface, backend validation, graphical presentation and interactive three-dimensional illustrations.

We are grateful to our classmates and families for their encouragement. Our work emphasizes a clear project scope, understandable explanations and a distinction between verified software behavior and health outcomes that have not been evaluated.

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Chapter 1

Problem Statement and Project Objectives

1.1 Project overview

CholesterolWise is a web-based educational application that explains manually entered LDL, HDL and total cholesterol readings. A React interface presents server-generated categories, charts and derived calculations. Interactive three-dimensional heart, DNA and PCSK9 illustrations connect the numerical report with the underlying physiology.

1.2 Problem statement

A laboratory report contains measurements and reference information, but a reader may still find it difficult to connect the numbers with cholesterol transport and clearance. The engineering problem is to provide a simple, consistent interface that explains these measurements, shows their reference categories and introduces the role of PCSK9 without implying that three measurements are sufficient for a medical diagnosis [1], [2].

1.3 Project objectives

- Accept LDL, HDL and total cholesterol in mg/dL with optional adult HDL reference context.
- Validate readings and classify them using explicit backend reference rules.
- Calculate non-HDL cholesterol and the total/HDL ratio using decimal arithmetic.
- Present categories, scales, charts and plain-language explanations on a dashboard.
- Explain LDL, HDL, cholesterol regulation and general lifestyle awareness.
- Demonstrate receptor recycling and PCSK9-associated degradation through interactive 3D scenes.
- Verify input handling, boundary rules, API reliability and interface behavior.

1.4 Scope and intended outcome

The implemented prototype covers five screens: Home, Analyzer, Dashboard, Learn and PCSK9. Its general reference categories apply to adults aged 20 and above. It does not provide personalized treatment targets, disease prediction, genetic testing or machine learning. Clinical interpretation requires wider health context [1], [2].

Chapter 2

Literature Survey and Proposed Solution

2.1 Review of related work

Reference categories and lipid interpretation. The NCEP Adult Treatment Panel III executive summary [3] documents the familiar LDL and total cholesterol category boundaries used by the prototype. These historical categories support transparent educational classification; they are not presented as a complete contemporary treatment algorithm. MedlinePlus [1], [2] provides accessible explanations and emphasizes the need for individual clinical context.

Discovery of the PCSK9 association. Abifadel et al. [4] identified PCSK9 mutations associated with autosomal dominant hypercholesterolemia. The study established a link between this gene and inherited elevated cholesterol. CholesterolWise uses this finding to explain why a gene can influence lipid regulation; the website does not infer a user's genotype from cholesterol values.

Human genetic evidence. Cohen et al. [5] examined LDL-lowering PCSK9 sequence variants and coronary events over a 15-year interval in the ARIC study. Lower LDL associated with these variants accompanied lower coronary disease incidence. This supports the biological relevance of the educational module, while providing no validation of this website as a risk calculator.

Receptor mechanism. Zhang et al. [6] studied PCSK9 binding to the LDL receptor and showed interference with receptor recycling, with increased receptor degradation. This mechanism informs the application's comparison between receptor recycling and the PCSK9-associated pathway. The scene is a schematic explanation rather than an atomic structure or quantitative simulation.

Clinical relevance of PCSK9. Sabatine et al. [7] reported the FOURIER randomized trial in 27,564 patients with established cardiovascular disease receiving statins. Evolocumab reduced LDL and cardiovascular events relative to placebo. The study explains the clinical importance of PCSK9 biology; its treatment findings do not establish any benefit from using an educational website.

2.2 Identified design opportunity

The project combines a transparent interpretation workflow with visual physiology education in one student-scale application. This is an integration objective, not a claim that existing products lack these features. Understanding and usability must be evaluated separately from the software's numerical correctness.

2.3 Evidence comparison

Source	Contribution to CholesterolWise	Limit of application
NCEP and MedlinePlus [1]-[3]	Explicit LDL, HDL and total reference explanations.	Categories are general; personal treatment targets vary.
Abifadel et al. [4]	Gene-to-protein introduction for PCSK9.	No genotype inference or genetic diagnosis.
Cohen et al. [5]	Connects inherited LDL regulation with human outcomes.	Genetic association is not a website prediction model.
Zhang et al. [6]	Recycling versus degradation animation.	Schematic geometry; no molecular dynamics.
Sabatine et al. [7]	Explains why PCSK9 is clinically relevant.	No medicine selection, dosing or treatment advice.

2.4 Proposed solution and implemented scope

A user enters readings from a laboratory report and selects an optional HDL context. React sends a JSON request to FastAPI. The backend validates the input, applies deterministic reference rules and returns the complete interpretation. The dashboard renders the response using result cards, a bar chart and calculated values.

The Learn section introduces cholesterol terminology and general lifestyle awareness. The PCSK9 section provides an animated DNA illustration and a stepwise comparison of LDL receptor pathways. Play, pause, stage selection and view-reset controls allow the learner to inspect the process at a comfortable pace.

2.5 Why a rule-based approach is suitable

The task is to explain established categories, so a readable ordered rule table is appropriate. It requires no training dataset, fitted parameters or prediction score. Rules can be inspected and tested directly at every cutoff. The same validated input produces the same interpretation and calculations.

2.6 Physiology and educational boundary

PCSK9 is a gene that encodes a protein involved in regulating LDL receptor availability. The website distinguishes the DNA illustration from the protein and receptor mechanism [6], [8]. Neither the animation nor the measured lipid values establish the presence of a PCSK9 mutation.

Chapter 3

System Design and Methodology

3.1 System architecture

The implementation separates the browser interface from the Python analysis service. The browser owns interaction and visual presentation; the server owns all reference classifications and derived calculations. No database is required for the current analysis session.

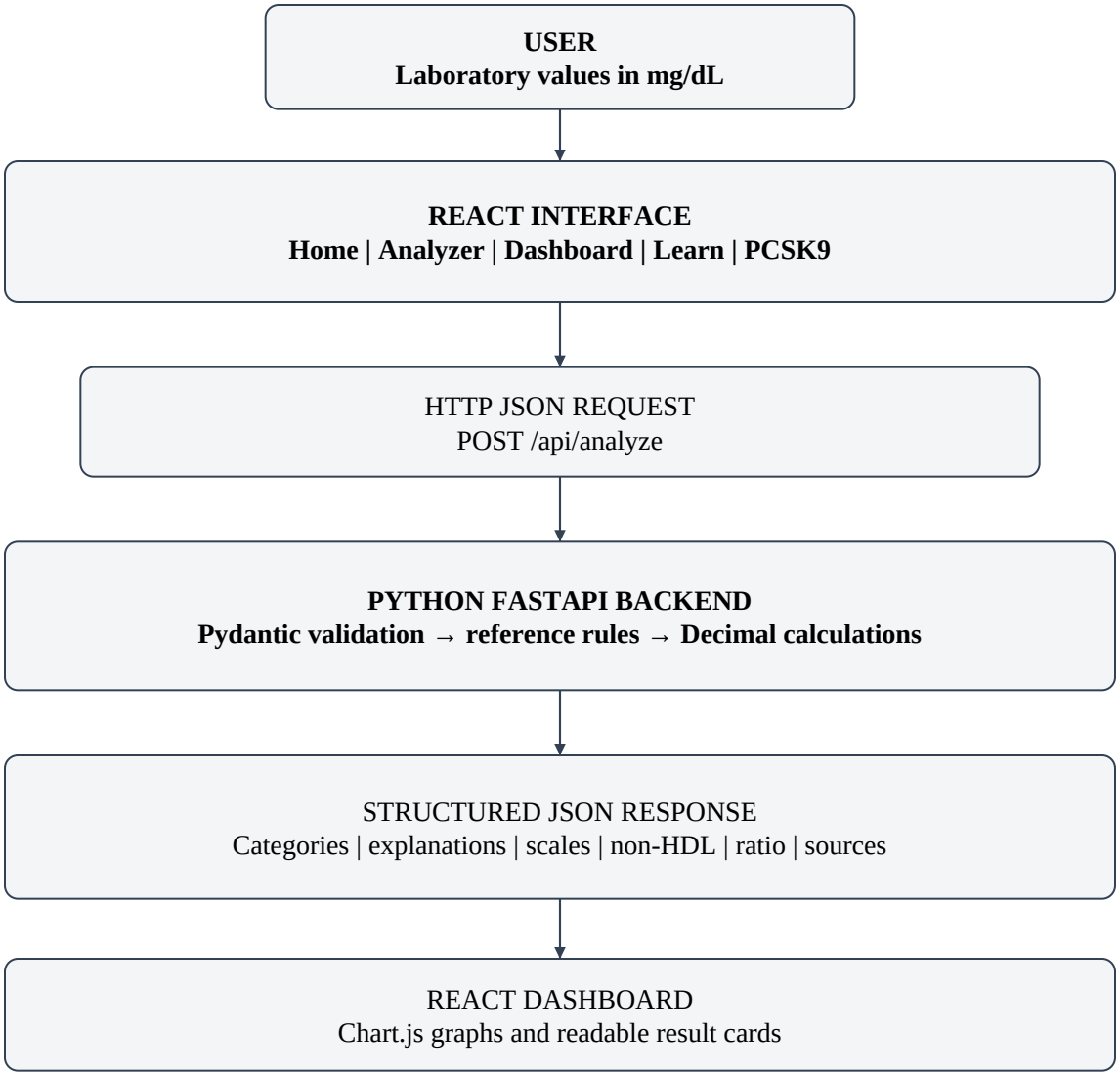


Figure 3.1 Implemented frontend and backend architecture

In the local production configuration, FastAPI serves the built React files and API from the same origin. Educational pages are frontend content; an analysis request requires the Python service to be available.

3.2 Input and analysis workflow

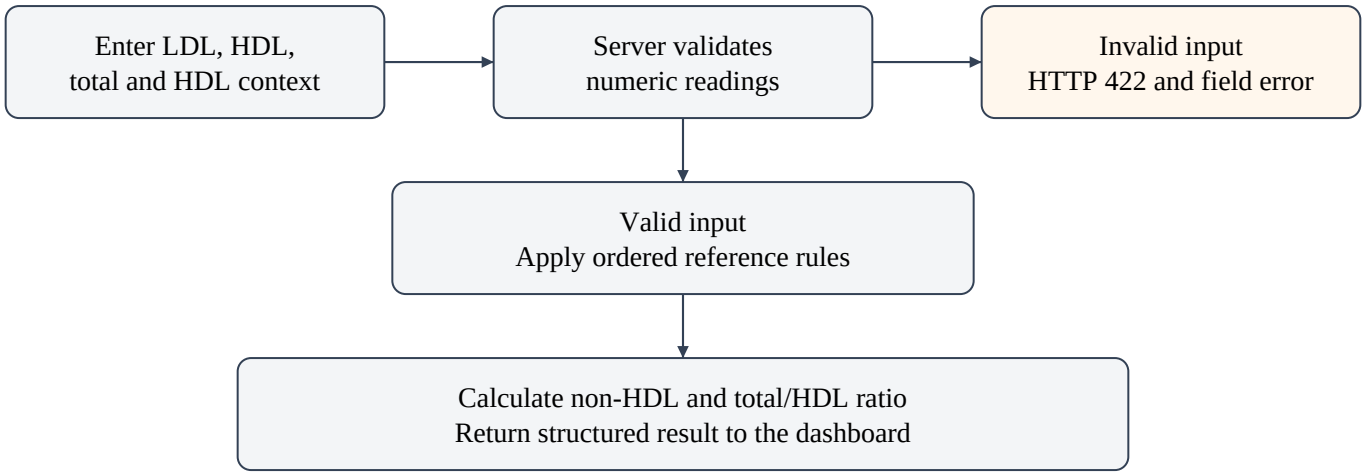


Figure 3.2 Valid input and validation-error paths

3.3 LDL and total cholesterol reference rules

Parameter	Category	Implemented interval in mg/dL
LDL	Optimal	0 < value < 100
LDL	Near optimal	100 <= value < 130
LDL	Borderline high	130 <= value < 160
LDL	High	160 <= value < 190
LDL	Very high	value >= 190
Total	Desirable	0 < value < 200
Total	Borderline high	200 <= value < 240
Total	High	value >= 240

Rules are checked in ascending order and the first matching interval is selected. Upper bounds are exclusive, so a decimal reading such as LDL 129.9 remains near optimal and 130 becomes borderline high. The conventional categories originate from general adult reference material [2], [3].

3.4 HDL context

HDL below 40 is low for the male context and below 50 is low for the female context. With context unspecified, 40 to below 50 is labelled context-dependent. Values above the applicable low threshold and below 60 are intermediate; values at least 60 are labelled generally favorable. A favorable HDL label does not cancel other risk factors [1], [2].

3.5 Exact backend calculation method

The analysis engine performs deterministic classification and two derived calculations. LDL is entered from the laboratory report; it is not estimated. Triglycerides are not collected, so the Friedewald equation is not used.

Derived measure	Formula and implementation
Non-HDL cholesterol	Total cholesterol - HDL cholesterol (mg/dL) [9]. Subtraction uses Python Decimal.
Total/HDL ratio	Total cholesterol / HDL cholesterol. Decimal division is rounded to two places using ROUND_HALF_UP. The ratio has no unit.

Worked example: total = 210.1 mg/dL and HDL = 45.2 mg/dL. Non-HDL = 210.1 - 45.2 = **164.9 mg/dL**. Total/HDL = 210.1 / 45.2 = 4.6482..., displayed as **4.65**. Decimal values are created from the validated readings' string representations to avoid common binary subtraction artifacts.

3.6 Server validation

Pydantic accepts finite positive JSON numbers and rejects strings, booleans, missing required fields and unexpected fields. Input caps are LDL 1000, HDL 300 and total 1500 mg/dL. These are typing and unit guardrails, not clinical thresholds. Total must not be lower than either entered LDL or entered HDL. The backend does not enforce a total-versus-component-sum rule.

3.7 API contract

Method and path	Purpose
POST /api/analyze	Validate readings; return categories, explanations, scales and calculations.
GET /api/reference-ranges	Expose reference categories, formulas, guardrails, sources and engine version.
GET /api/health	Return service health and analysis engine version.

The analysis response includes readings, units, metric results, non-HDL, ratio, a discussion prompt, source URLs and engine version 1.0.0. Invalid requests return HTTP 422 with field-level messages. React displays these messages beside the relevant input.

3.8 PCSK9 visualization methodology

The educational scene contrasts the reuse of an LDL receptor with PCSK9-associated receptor degradation [6], [8]. The schematic uses Three.js geometry through React Three Fiber. It is intended to communicate a sequence of events, not protein shape, binding affinity or a patient-specific rate.

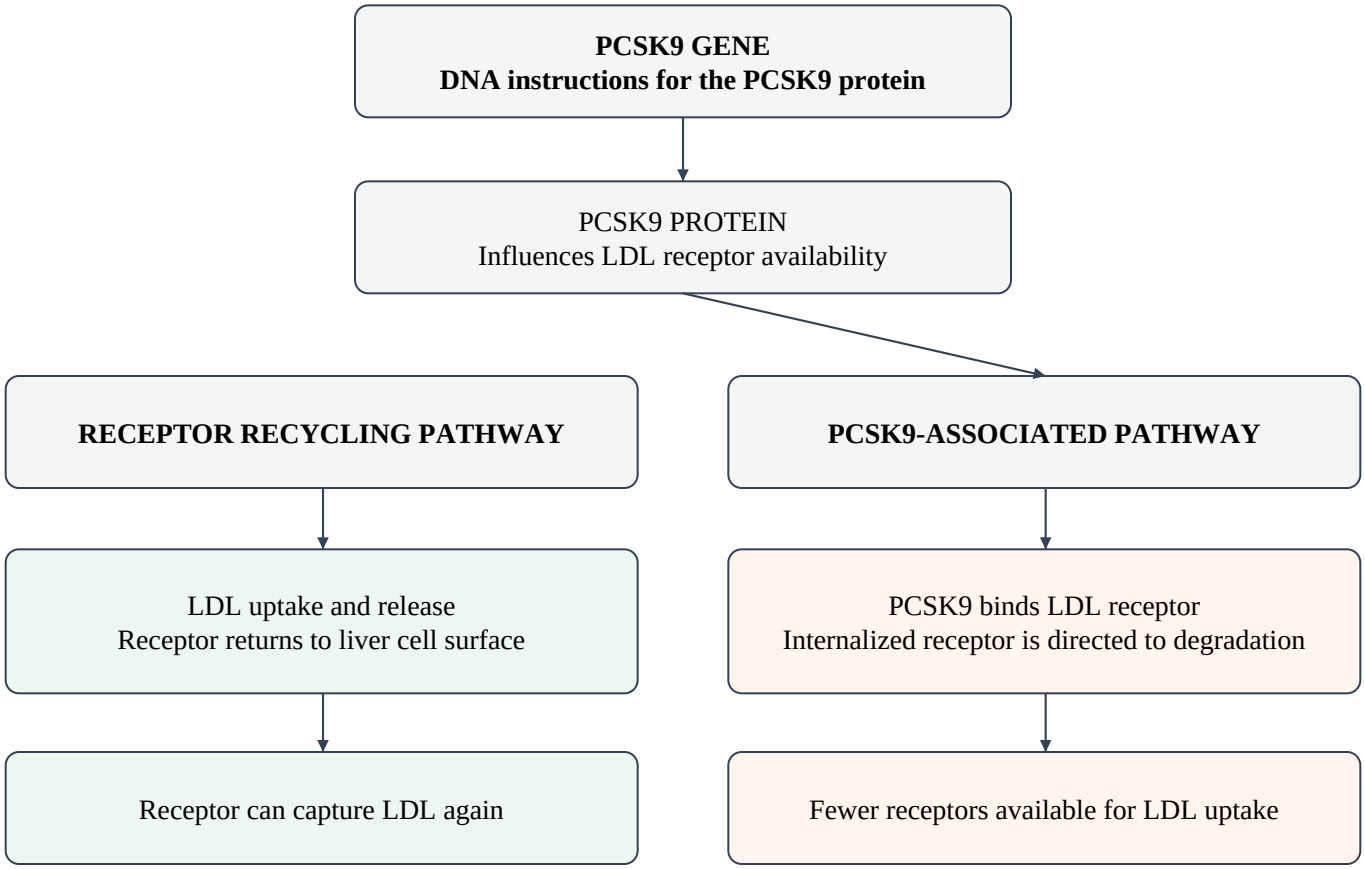


Figure 3.3 Schematic PCSK9 and LDL receptor pathways [6], [8]

3.9 Interaction and accessibility

Users can rotate the scenes, pause animation, choose explanation stages and reset the view. The interface supports reduced-motion preferences and a two-dimensional fallback when WebGL is unavailable. Responsive layouts, visible labels and keyboard-accessible controls support use across screen sizes.

3.10 Data handling

Readings are processed in server memory and retained only in the active React session. The app implements no accounts, patient database or request-body logging, and API responses use Cache-Control: no-store. Reloading clears the current results. A future remote deployment must also configure its hosting and proxy logs appropriately.

Chapter 4

Implementation, Experimentation and Results

4.1 Implemented technology stack

Layer	Technology	Purpose
Frontend	React 19, TypeScript, Vite	Component-based pages and production build [10].
Styling and navigation	Custom CSS; React Router	Navy landing theme, light dashboard and responsive navigation.
Interactive 3D	Three.js; React Three Fiber; Drei	Heart, DNA and receptor illustrations with camera controls [12].
Charts	Chart.js; react-chartjs-2	Lipid values and result visualization [13].
Backend	Python; FastAPI; Pydantic; Uvicorn	Typed API, strict validation and server execution [11].
Calculation	Python Decimal; ordered rules	Reference-range comparison and precise derived arithmetic.
Verification	pytest; HTTPX; Vitest	Backend tests and frontend form/API tests.

4.2 Main application modules

Home: project introduction, animated heart and navigation to analysis.
Analyzer: numeric lipid inputs, optional HDL context and field errors.
Dashboard: categories, reference scales, chart, explanations and derived values.
Learn: cholesterol concepts and general health-awareness information.
PCSK9: DNA illustration and interactive receptor-pathway comparison.

4.3 Implementation sequence

Development proceeded through the interface and core routes, interactive 3D scenes, rule-based interpretation, Python API integration and verification. Clinical rules were centralized in the backend, while React retained presentation and basic input checks. The API-dependent version has been verified locally; public full-stack deployment requires a Python-capable host.

4.4 Implementation workflow

A single request pipeline connects the form to the result screen. The frontend does not silently recompute clinical results if the server is unavailable. A failed request produces a visible message while preserving the entered values.

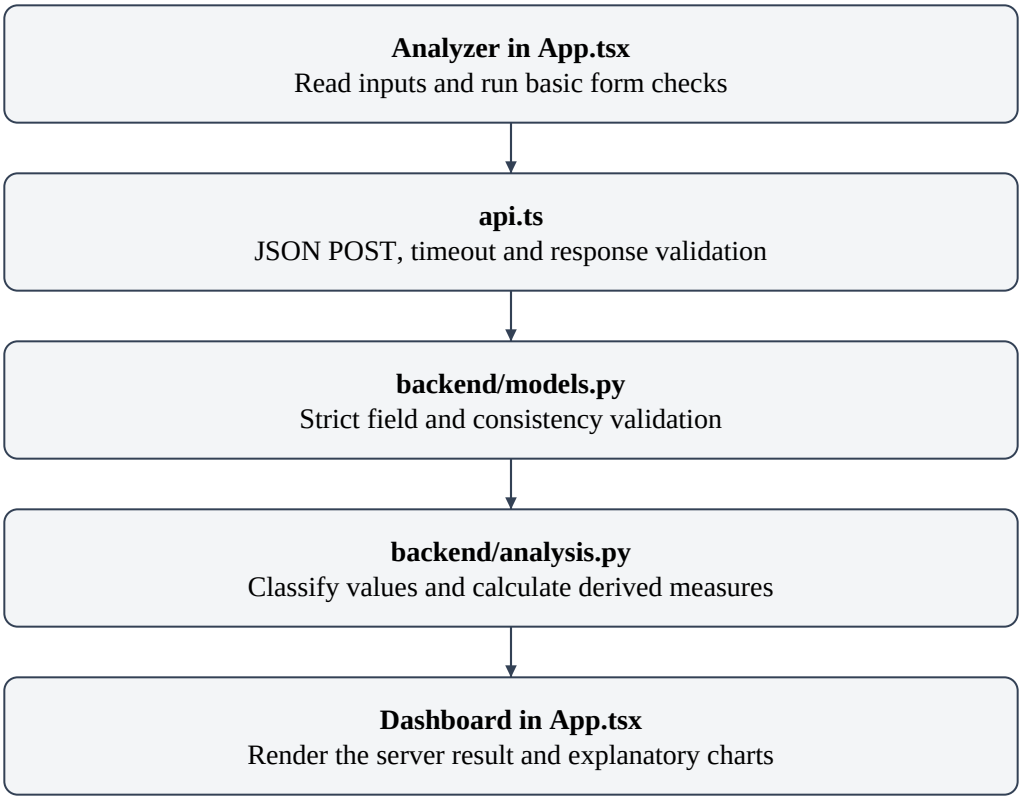


Figure 4.1 Implemented request and rendering pipeline

4.5 Experimental setup and procedure

Verification used the local Windows development environment, the production React build, FastAPI TestClient and browser checks. Test inputs are synthetic lipid values rather than patient records. Tests compare actual returned labels and arithmetic with explicit expected outputs, including values immediately below and at category boundaries.

The API client has a 15-second timeout, supports cancellation and rejects invalid or incomplete responses. Browser checks covered sample analysis, field-error display, decimal results, mobile navigation and 3D controls. No volunteer usability study or clinical evaluation was conducted.

4.6 Test inputs and observed calculations

The cases below exercise the implemented rules. All values are illustrative software-test inputs in mg/dL. HDL context is unspecified unless shown otherwise. The ratio column uses backend half-up rounding to two decimal places.

Case	LDL / HDL / Total	LDL; HDL; total categories	Non-HDL	Ratio
1	80 / 65 / 170	Optimal; generally favorable; desirable	105	2.62
2	120 / 45 / 210	Near optimal; context-dependent; borderline high	165	4.67
3	150 / 35 / 230	Borderline high; low; borderline high	195	6.57
4	195 / 38 / 260	Very high; low; high	222	6.84
5 Female	120 / 45.2 / 210.1	Near optimal; low; borderline high	164.9	4.65

4.7 Recorded software verification

Check	Recorded result	Coverage
Backend automated suite	46 passed	Category boundaries, HDL context, validation, decimal results, API behavior and request isolation.
Frontend automated suite	16 passed	Nine input-validation tests and seven API-client tests.
Production build	Passed	TypeScript checking and Vite production bundling.
Browser checks	Completed	Sample flow, server errors, decimal display, responsive navigation and 3D interaction.

4.8 Interpretation of results

The recorded 62 passing automated tests support agreement with the implemented rules for the tested cases. They do not establish diagnostic accuracy, comprehensive test coverage or an improvement in health awareness. The large 3D bundle remains a performance consideration; low-powered devices and slow connections need broader evaluation.

The results demonstrate a functioning input-to-dashboard workflow and reproducible calculations. Usability, learning outcomes, clinical content review and public hosting remain separate evaluation or deployment tasks.

4.9 Project screenshots

The following figures show the implemented website. They document the actual interface and use illustrative readings. They are not conceptual mockups or patient reports.

4.9.1 Landing page

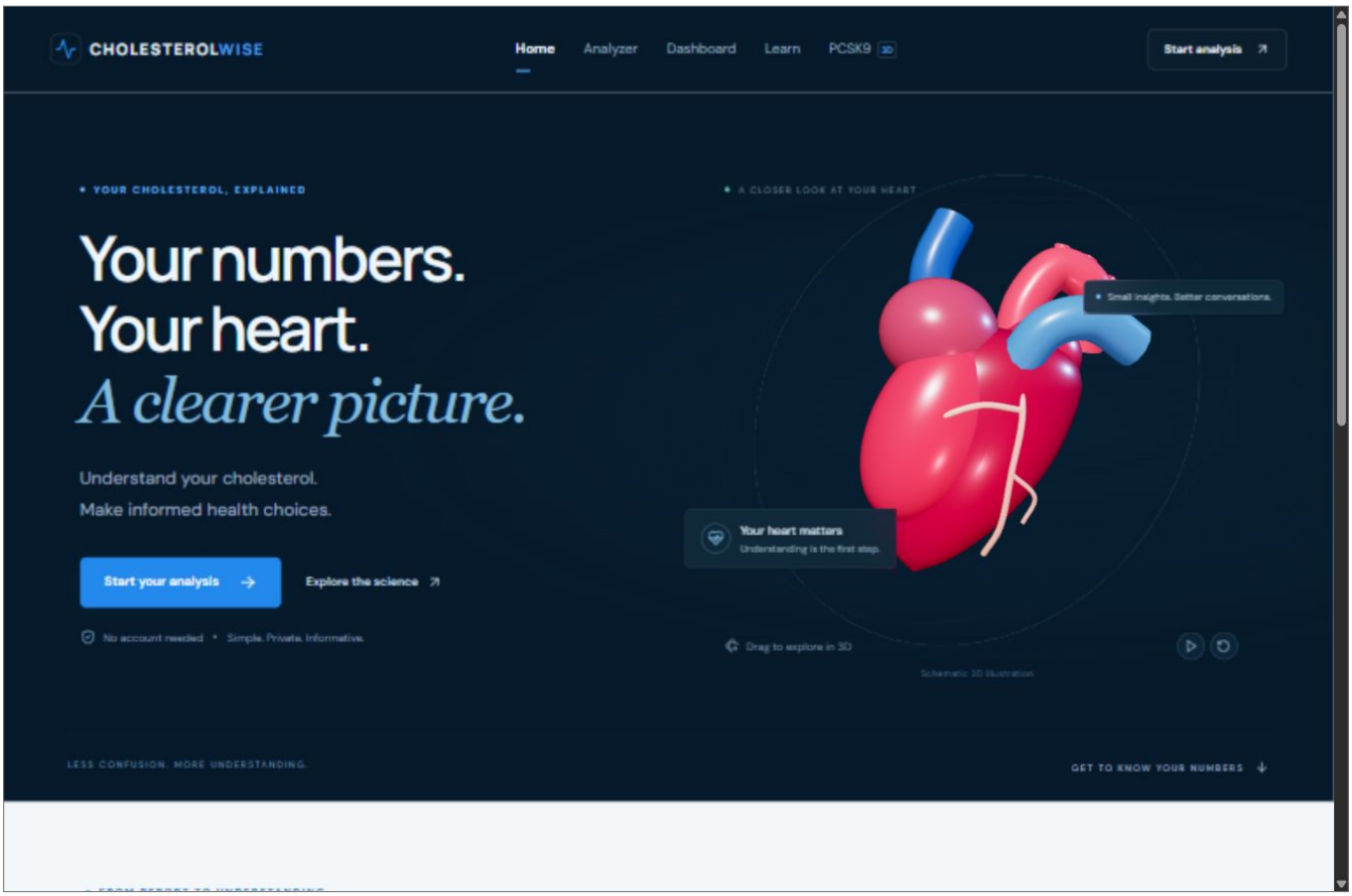


Figure 4.2 CholesterolWise landing page with interactive schematic heart

The landing page uses a dark navy background, blue accents and a prominent route to the analyzer. The heart scene introduces the subject visually and supports drag-to-rotate, pause and reset interactions.

4.9.2 Educational integration

Navigation connects the landing page with the analyzer, dashboard, cholesterol learning content and PCSK9 module. The 3D objects are labelled as schematic illustrations. Their role is explanatory; they do not visualize a user's heart or genetic data.

4.9.3 Backend-connected dashboard

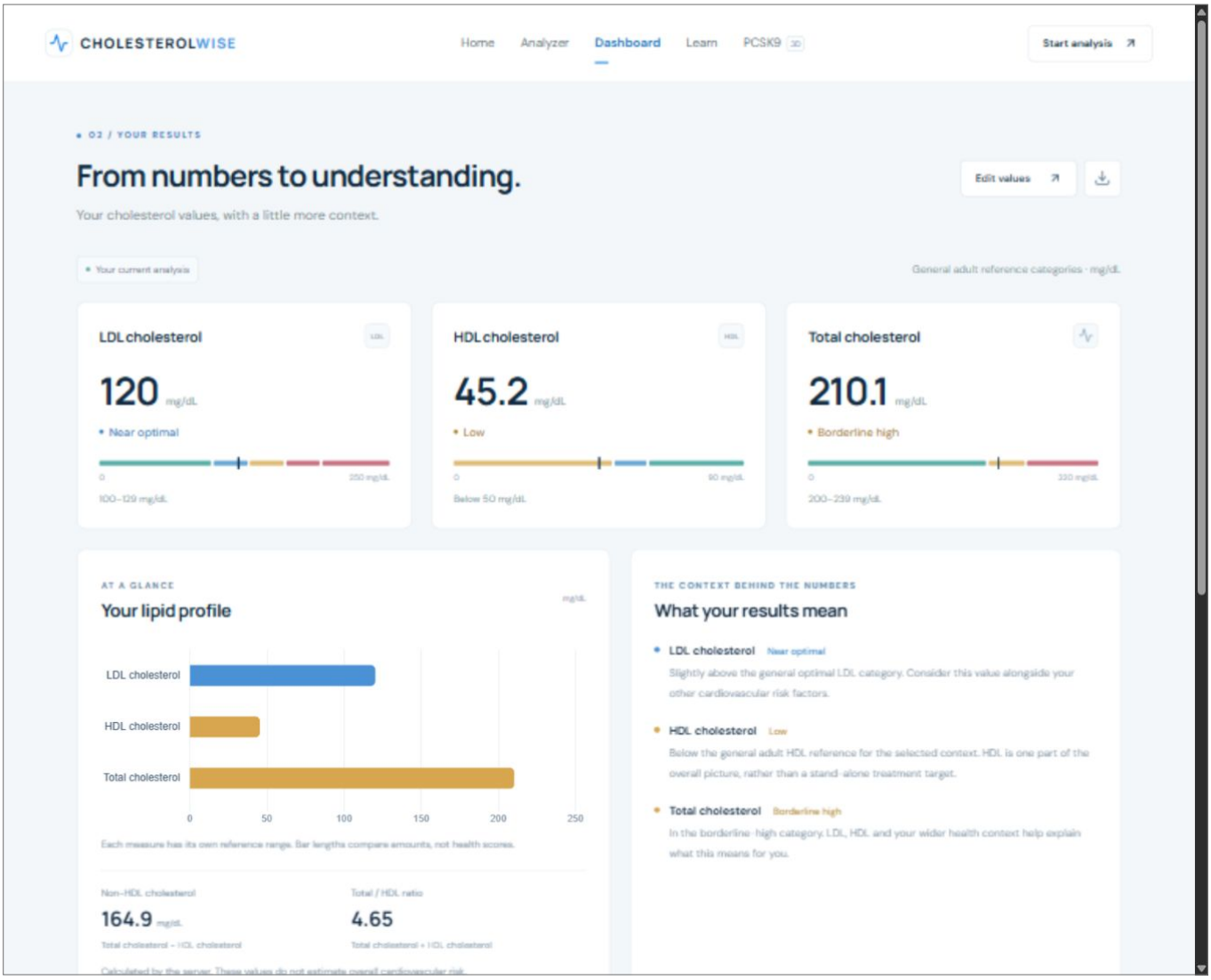


Figure 4.3 Dashboard for LDL 120, HDL 45.2 and total 210.1 mg/dL, female context

The dashboard shows near-optimal LDL, low HDL for the selected female context and borderline-high total cholesterol. Its non-HDL result of 164.9 mg/dL and total/HDL ratio of 4.65 come from the Python backend. The chart compares quantities; bar length is not a health score.

4.10 Demonstration and deployment status

The project can be demonstrated locally using the supplied PowerShell launcher, which builds React and starts FastAPI. The current private static preview cannot execute the Python backend. The complete application needs a Python or container host before the integrated analysis can be offered remotely.

Chapter 5

Conclusion and Future Enhancements

5.1 Conclusion

CholesterolWise demonstrates how a web application can combine transparent numerical interpretation with visual biomedical education. The implemented React interface accepts lipid readings and renders the results returned by a Python FastAPI service. Reference classifications, non-HDL cholesterol and the total/HDL ratio are calculated in one backend module.

Interactive heart, DNA and PCSK9 scenes connect the user interface with cholesterol physiology. The literature supports the biological explanation, while the software checks establish the tested behavior of the application. The project therefore provides a working educational prototype without claiming disease prediction, treatment effectiveness or measured improvement in user knowledge.

5.2 Future enhancements

- Add authenticated profiles and optional history, with consent, deletion controls and secure storage.
- Support longitudinal charts and downloadable summaries of user-selected readings.
- Introduce report upload with user confirmation of extracted values before analysis.
- Add carefully validated unit conversion and clearer support for laboratory-specific reference information.
- Expand Hindi and Marathi content, keyboard testing and accessibility evaluation.
- Improve 3D loading performance and evaluate the interface on a wider range of mobile devices.
- Arrange professional review of educational content and conduct a small usability study.
- Extend PCSK9 education with reviewed explanations of genetic variation, keeping it separate from individual genetic assessment.

5.3 Limitations and learning outcomes

The prototype uses general adult references and collects only a limited set of lipid readings. It does not incorporate full medical history, blood pressure, medications or other clinical factors. Persistent records, report extraction and public backend hosting are not completed features. The work supports learning in React development, typed APIs, validation, decimal arithmetic, testing, 3D interaction and responsible use of biomedical evidence.

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Online resources verified on November 2, 2026. Project source: supplied CholesterolWise brief and the implemented React/FastAPI codebase, analysis engine 1.0.0. Institutional layout follows the supplied synopsis reference.